

CEN 352 – Final Project Report

TAIRA: Traffic Accident Intelligent Risk Advisor

Course: CEN 352 – Artificial Intelligence

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1. Introduction

Traffic accidents arise from interacting environmental, roadway, and human factors. Predicting potential injury severity under different conditions can support analysis, planning, and awareness. This project develops **TAIRA** — **Traffic Accident Intelligent Risk Advisor**, a hybrid intelligent agent that combines:

1. **A Random Forest classifier** to predict the *Most Severe Injury* category.
2. **A rule-based expert system** producing human-interpretable risk levels (LOW–CRITICAL).
3. **An interactive Streamlit dashboard** for scenario exploration and crash-pattern visualization.

TAIRA demonstrates key CEN 352 concepts, including intelligent agent design, supervised learning, rule-based reasoning, evaluation metrics, and ethical reflection.

2. Problem Definition Using the PEAS Framework

Performance Measures

- Accuracy and weighted F1-score of injury prediction
- Clarity and consistency of rule-based risk explanations
- UI usability, responsiveness, and interpretability

Environment

Chicago traffic crash data containing categorical features such as:

- Weather, lighting, road surface
- Traffic control device
- Crash type & contributory cause
- Injury outcomes

Actuators

- Predicted injury severity

- Assigned risk level
- Natural-language explanation
- Interactive plots and historical summaries

Sensors

- User dropdown inputs
- Encoded dataset features
- Outputs from the trained Random Forest model

3. Dataset Summary and Preprocessing

The dataset includes tens of thousands of Chicago crash records. The target variable **Most Severe Injury** consists of five ordered classes:
NO INDICATION → REPORTED → NON-INCAPACITATING → INCAPACITATING → FATAL.

Preprocessing steps

- Remove rows with missing critical fields
- Convert all fields to uppercase strings
- Ensure dropdown values match model categories
- Use OneHotEncoding for all inputs (categorical-only requirement)

Because severe injuries are rare, the dataset is imbalanced; evaluation uses weighted metrics.

4. Machine Learning Component

A **Random Forest Classifier** was selected due to its stability on categorical expansions, nonlinear interactions, and robustness to noise.

Pipeline

- 80/20 train–test split
- OneHotEncoder → RandomForest
- Class weights balanced
- Metrics: accuracy + weighted F1

Results

- **Accuracy:** ~0.71–0.76
- **Weighted F1:** ~0.68–0.72

5. Rule-Based Reasoning Layer

To provide interpretable safety guidance, TAIRA includes an expert system evaluating hazardous combinations.

Examples of Rules

CRITICAL:

- ML predicts *Fatal* or *Incapacitating*
- High-risk clusters (e.g., ICE + DARKNESS + NO CONTROLS; HEAD-ON + poor visibility)

HIGH:

- ML predicts *Non-incapacitating*
- Snow/ice/freezing rain
- Darkness with no control
- Hazardous crash types (Head-on, Angle)

MEDIUM:

- Wet/slushy surface, dawn/dusk, moderate causes

LOW:

- All other conditions

This layer strongly improves interpretability and human-centered decision logic.

6. Hybrid Agent Architecture

TAIRA integrates both techniques through a structured workflow:

1. User inputs (dropdowns) → feature dictionary
2. Data transformed to ML format
3. Random Forest predicts injury severity
4. Rule system evaluates environmental hazards
5. Final output:
 - predicted injury category
 - risk level
 - a plain-language explanation

This satisfies the requirement for multi-technique AI integration

7. Streamlit Dashboard

TAIRA is deployed as a modern multi-page web interface.

Home

- Overview of TAIRA
- Gradient dark theme + interactive navigation cards

Interactive Lab

1. **Pre-Crash Risk Tool**
 - Dropdown-only scenario input
 - ML prediction + rule-based risk level
 - Explanation text
2. **Historical Scenario Lookup**
 - Average injuries for chosen cause + crash type
 - Most common damage category

Crash Patterns & Story

- Monthly, hourly, and weekday crash distributions
- Weather, lighting, and surface condition charts
- Auto-generated narrative summarizing insights

About TAIRA

- PEAS framework summary
 - Architecture + evaluation
 - Ethical notes and limitations
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8. Evaluation

Model

- Strong performance on majority classes
- Expected difficulty predicting rare fatal cases

Rule System

- Enhances interpretability
- Reinforces critical hazards
- Provides consistent causal explanations

UI Usability

- Dropdowns make the tool accessible
- Dashboard layout is clear and informative
- Explanations guide understanding of model behavior

9. Ethical Considerations

TAIRA is an educational tool, not a real-time crash-risk predictor. Potential risks:

- **Automation bias:** Users may overtrust predictions
- **Dataset bias:** Uneven crash reporting, location imbalance
- **Missing variables:** Driver behavior, vehicle speed, distractions not captured

The app warns users about these limitations on the About page.

10. Conclusion

TAIRA successfully demonstrates a hybrid AI system combining machine learning and rule-based reasoning within an interactive, professionally styled dashboard. Core achievements include:

- Working Random Forest injury severity predictor
- Custom expert system for environmental risk reasoning
- Multi-page Streamlit application with modern UI
- Strong evaluation using Accuracy and F1
- Ethical reflection integrated into system design

The project thoroughly applies CEN 352 concepts including intelligent agents, supervised learning, rule-based inference, evaluation, and interface development.